

2020

Advanced Computing Program

Entrance Exam

March 14, 2020

Problem S1: Counting

Time limit: 1.0 second(s)

Memory limit: 512 MiB

Description

A message only consists of the lowercase **a**, **b**, and **c**. Could you count how many times each letter (from **a** to **c**) appears in the message?

Input Format

Read from standard input.

The first line of each input contains a string, which means the message.

Output Format

Write to the standard output.

Output only one integer per line. The first line is how many times **a** appears in the message. Results of **b** and **c** in the second and third lines.

Sample Input

```
1 aaabbc
```

Sample Output

```
1 3
2 2
3 1
```

Subtasks

Task	Length of the message	Is the message contains only the same letters?
1,2,3	$0 < \text{length} \leq 10^4$	Yes
4,5,6,7,8,9,10		No

Problem S2: Covering

Time limit: 2.0 second(s)

Memory limit: 512 MiB

Description

An **integer interval** $[l_a, r_a]$, $l_a \leq r_a$, is a set of all consecutive integers beginning with l_a and ending with r_a . You can merge some intervals into a larger interval. For example, $[2, 4]$, $[3, 5]$ and $[6, 7]$ can be merged into $[2, 7]$. We define that $[l_a, r_a]$ **covers** $[l_b, r_b]$ only if $l_a \leq l_b \wedge r_b \leq r_a$. For example, $[2, 6]$ **covers** $[2, 4]$ and $[4, 5]$, however, $[2, 6]$ does **not cover** $[3, 7]$.

There are n **integer intervals** $[l_i, r_i]$. To **cover** $[l_t, r_t]$, what is the minimum number of intervals to be selected and merged?

Input Format

Read from standard input.

The first line of each input is an integer T , which means the input consists of T cases.

The first line for each case contains three integers, n , l_t and r_t . Each of the next n lines has two integers giving l_i and r_i .

Output Format

Write to the standard output.

For each test case output the minimum number of intervals needed. If it is impossible, output -1.

Sample Input

```
1 2
2 5 5 10
3 1 3
4 3 6
5 5 7
6 7 9
7 8 11
8 3 5 9
```

9	1	3
10	4	8
11	2	3

Sample Output

1	2
2	-1

Sample Explanation

There are two test cases. For the first case, [5, 7] and [8, 11] can cover [5, 10]. For the second cases, it is impossible to cover [5, 9].

Subtasks

Task	T	n	$ l_i , r_i , l_t , r_t $
1,2	= 20	≤ 10	$\leq 10^9$
3,4		$\leq 10^3$	$\leq 10^3$
5,6			$\leq 10^9$
7,8,9,10		$\leq 10^5$	

Problem S3: Typing

Time limit: 1.0 second(s)

Memory limit: 512 MiB

Description

Your boss asks you to type exactly n letter **B** on the notepad. You are surprised to find that your keyboard now has only **Ctrl**, **A**, **C**, and **V** keys available. Fortunately, one character **B** is present on your notepad.

With Ctrl, A, C, and V, you can only perform one of the following two operations for each step:

- 1 **Copy:** Copy all the characters from the notepad. You can only copy all characters.
- 2 **Paste:** Paste the characters which are copied last time. Pasted characters are appended to the notepad.

What is the minimum number of steps to get n letter **B**?

Input Format

Read from standard input.

The first line of each input is an integer T , which means the input consists of T cases.

Each case contains only one integer, n .

Output Format

Write to the standard output.

For each test case output the minimum number of steps to get n letter **B**. If it is impossible, output -1.

Sample Input

```
1 3
2 1
3 2
4 4
```

Sample Output

```
1 0
2 2
3 4
```

Sample Explanation

There are three test cases.

For the first case, $n = 1$. We get one **B** intially.

For the second case, $n = 2$. We get one **B** intially. In step 1, we perform **Copy**. In step 2, we perform **Paste** and get **BB**.

For the third case, $n = 4$. We get one **B** intially. In step 1, we perform **Copy**. In step 2, we perform **Paste** and get **BB**. In step 3, we perform **Copy**. In step 2, we perform **Paste** and get **BBBB**.

Subtasks

Task	T	n
1,2,3	= 20	$\leq 10^3$
4,5,6		$\leq 10^6$
7,8,9,10		$\leq 10^9$